



- Faculty of Engineering  
Computer Department



# ST Project Submission

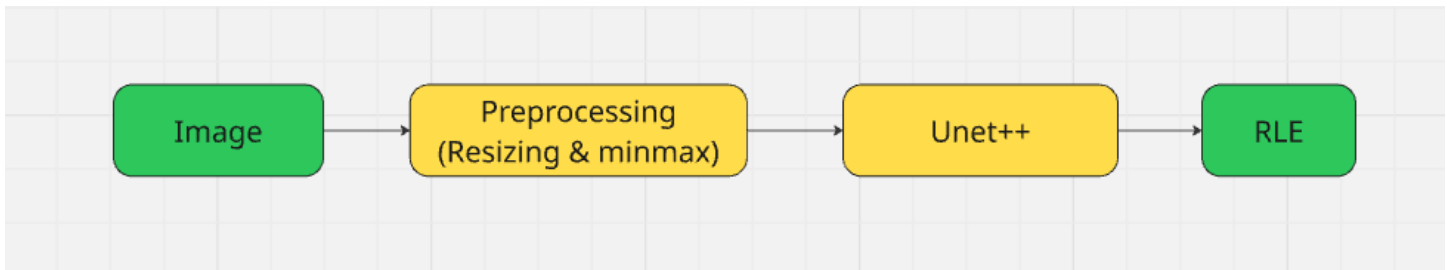
## Team 11

### Team Members

| Num | Full Name in ARABIC       | SEC | BN |
|-----|---------------------------|-----|----|
| 1   | مروان سامي محمد           | 2   | 18 |
| 2   | مروان عماد مصطفى          | 2   | 19 |
| 3   | سيف الدين هاني عبد الفتاح | 1   | 20 |
| 4   | محمد عاطف احمد            | 2   | 11 |



## Project Pipeline



1. First, the Images are resized from  $512 \times 512$  to  $256 \times 256$  to reduce the number of parameters and increase the model speed (Some training examples were removed, which is further explained in the EDA section).
2. Min-max normalization is then used for each channel to avoid having a single channel dominating if it has a higher range of values.
3. Unet++ is the Deep Learning architecture we used, as it proves to be a robust model for medical applications.
4. The output of the Unet++ is a  $256 \times 256$  mask that's then thresholded to be binary.
5. Lastly, RLE encoding is used to compress the output mask.



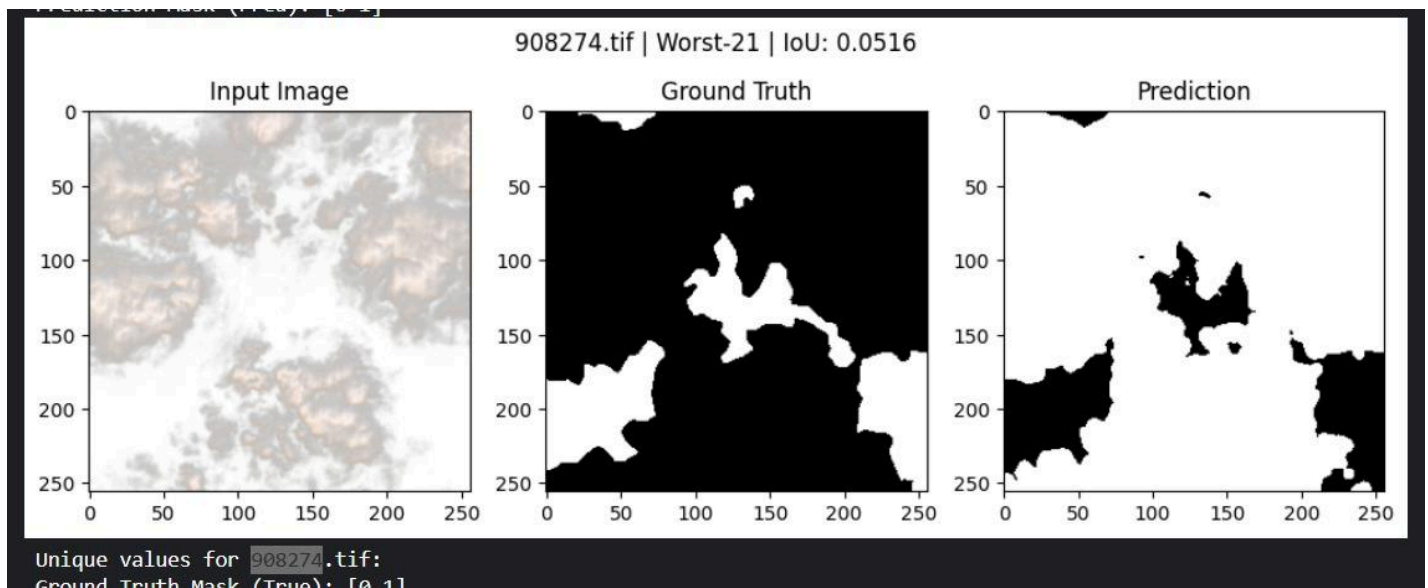
## Preprocessing Module

To reduce the number of parameters and the size of the model, we resized the model to  $256 * 256$  instead of  $512 * 512$ . This also helped to reduce the training time.

To remove the effect of some bands dominating the training due to them having greater ranges of values compared to other bands, we used min-max normalization for each band so each band has values ranging from 0 to 1.

## Exploratory Data Analysis

We noticed that there were a lot of mislabeled images, such as masks with all zeros. masks with inverted pixels, so we used the trained model to detect these masks by detecting the masks that had very little intersection with the predicted mask. Also, some of the images were filtered by the human eye.





## Model Selection & Training Module

As we needed to do a pixel-wise classification, we needed to use a model with an architecture like U-Net, and we actually tried it and got a score of 75%, which made us eager to find a better solution with a higher score.

We found papers talking about U-Net++ and its usage in medical applications, which implies its robustness as medical applications are very strict.

After using U-Net++, we got much better results, but still the score wasn't up to our expectations so we investigated the misclassified images, and to our surprise, we found out that the training (ground truth) masks had some errors, with some masks showing only zeros while the whole image has clouds.

We tested with upsampling layers and transpose convolution layers, as upsampling layers reduced the number of parameters and the size of the model, but reduced the model accuracy. The transpose convolution layers gave slightly higher accuracy but larger model size and number of parameters. The final decision was to use upsampling layers, as the accuracy tradeoff wasn't really worth it.

Both models were trained with different epochs and learning rates, which is further explained in the following section.



## Performance Analysis Module

| Model Name | Type             | Parameters                   | Accuracy         | Training Size | Test Size              |
|------------|------------------|------------------------------|------------------|---------------|------------------------|
| RF PCA     | Machine Learning | 15 MaxDepth<br>100 estimator | 87.39%<br>87.82% | 1200<br>1200  | 6000<br>Full Test Data |
| Unet++     | Deep Learning    | 25 epochs<br>lr = 0.001      | 91.66%           | 7000          | 1500                   |
| Unet++     | Deep Learning    | 35 epochs<br>lr = 0.0001     | 91.48%           | 8000          | 1000                   |



## Enhancements and Future Work

After the low accuracy we got in the competition, we concluded that the model we used was overfitted to the training data, so we should have used a simpler model to have better generalization, like Unet, and add dropout layers while increasing the number of epochs, which can help with the overfitting problem we faced.

Maybe we could also try other model architectures from other research papers, as the one we used was [the Unet++ research paper](#).

## Workload Distribution

| id      | Workload                                    |
|---------|---------------------------------------------|
| 9211121 | Machine learning part                       |
| 9211123 | image filtration, preprocessing, validation |
| 9210512 | Model selection and implementation          |
| 9211010 | Model selection and implementation          |